

ARPAN MUKHERJEE

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Education

- 2019–2024 **Rensselaer Polytechnic Institute.**
- Ph.D. candidate, department of Electrical Computer & Systems Engineering
 - GPA: 3.95/4
 - Advisor: Ali Tajer
- 2017–2019 : **Indian Institute of Technology, Kharagpur.**
- M.Tech., Department of Electronics and Electrical Communication Engineering
 - GPA: 9.19/10
 - Advisor: Mrityunjoy Chakraborty
- 2013–2017 : **West Bengal University of Technology.**
- B.Tech., Department of Electronics and Communication Engineering
 - GPA: 9.05/10
 - Advisor: Krishanu Datta

Work Experience

- 12.24 – present **Imperial College London, London.**
- Informed-AI postdoctoral research associate
 - Advisor: Deniz Gündüz
- 06.24 – 08.24 **IBM Research, NY.**
- Research Internship
 - Project: Bandit-based Exploration for Prompt Engineering in LLMs
 - Collaborators: Djallel Bouneffouf, Miao Liu
 - Manager: Prasanna Sattigeri
- 06.23 – 08.23 **IBM Research, NY.**
- Research Internship
 - Project: Group testing for combinatorial bandits
 - Collaborators: Shashanka Ubaru, Keerthiram Murugesan and Karthikeyan Shanmugam
 - Manager: Lior Horesh
- 06.21 – 08.21 **IBM Research, NY.**
- Research Internship
 - Project: Data-aware client selection in federated learning
 - Collaborators: Theodoros Salonidis, Shiqiang Wang and Georgios Kollias
 - Manager: Theodoros Salonidis

Publications

Preprints

- 2025 **SharedRep-RLHF: A Shared Representation Approach to RLHF with Diverse Preferences, (working paper).**
A. Mukherjee, M. Bullo and D. Gündüz
- 2025 **Preference-centric Bandits: Optimality of Mixtures and Regret-efficient Algorithms.**
M. Tatli, A. Mukherjee, Prashanth LA, K. Shanmugam and A. Tajer

2024 **Combinatorial Multi-armed Bandits: Arm Selection via Group Testing.**
A. Mukherjee, S. Ubaru, K. Murugesan, K. Shanmugam and A. Tajer

Journals

T-IT 2024 **Best Arm Identification in Stochastic Bandits: Beyond β -optimality**, *IEEE Transaction on Information Theory* (accepted for publication), November 2024.
A. Mukherjee and A. Tajer

T-IT 2024 **Optimal Best Arm Identification with Fixed Confidence in Restless Bandits**, *IEEE Transaction on Information Theory* (accepted for publication), June 2024.
P. N. Karthik, V. Y. F. Tan, A. Mukherjee and A. Tajer

JSAIT 2024 **Robust Causal Bandits for Linear Time-varying Models**, *IEEE Journal on Selected Areas in Information Theory* (accepted for publication), February 2024.
Z. Yan, A. Mukherjee, B. Varici and A. Tajer

JSAIT 2023 **SPRT-based Efficient Best Arm Identification in Stochastic Bandits**, *IEEE Journal on Selected Areas in Information Theory*, vol. 4, pp. 128-143, July 2023.
A. Mukherjee and A. Tajer

TSP 2022 **Active Sampling of Multiple Sources for Sequential Estimation**, *IEEE Transactions on Signal Processing*, vol. 70, pp.4571-4585, July 2022.
A. Mukherjee and A. Tajer

TSP 2020 **ImdLMS: An Imputation based LMS algorithm for Linear System Identification with Missing Input Data**, *IEEE Transactions on Signal Processing*, vol. 68, pp. 2370-2385, 2020.
S. Mukhopadhyay and A. Mukherjee

Conferences (* denotes equal contribution)

AISTATS 2025 **Risk-sensitive Bandits: Arm Mixture Optimality and Regret-efficient Algorithms**, *Proc. International Conference on Artificial Intelligence and Statistics*, May 2025.
M. Tatli*, A. Mukherjee*, Prashanth LA, K. Shanmugam and A. Tajer

ISIT 2024 **BAI in Exponential Family: Efficiency and Optimality**, *Proc. International Symposium on Information Theory*, July 2024.
A. Mukherjee and A. Tajer

ISIT 2024 **Improved Bound for Robust Causal Bandits with Linear Models**, *Proc. International Symposium on Information Theory*, accepted for publication.
Z. Yan, A. Mukherjee, B. Varici and A. Tajer

ISIT 2022 **SPRT-based Best Arm Identification in Stochastic Bandits**, *Proc. International Symposium on Information Theory*, Helsinki, Finland, June 2022.
A. Mukherjee and A. Tajer

NeurIPS 2021 **Mean-based Best Arm Identification in Stochastic Bandits under Reward Contamination**, *Proc. Advances in Neural Information Processing Systems*, virtual, December 2021.
A. Mukherjee, A. Tajer, P. Das and P.-Y. Chen

ISIT 2021 **Active Binary Classification of Random Fields**, *Proc. International Symposium on Information Theory*, Melbourne, Australia, July 2021.
A. Mukherjee, A. Tajer, P. Das and P.-Y. Chen

ICASSP 2021 **Active Estimation from Multimodal Data**, *Proc. International Conference on Acoustics, Speech and Signal Processing*, Toronto, Canada, June 2021.
A. Mukherjee, A. Tajer, P. Das and P.-Y. Chen

Teaching Experiences

- Spring 2024 **Teaching Assistant**, *Electronic Instrumentation*, (ENGR 2300), RPI.
Fall 2023 **Teaching Assistant**, *Electronic Instrumentation*, (ENGR 2300), RPI.
Fall 2022 **Teaching Assistant**, *Electronic Instrumentation*, (ENGR 2300), RPI.

Fellowships & Awards

- 2025 **Allen B. Dumont** (RPI ECSE doctoral dissertation award)
2022 **Winner** of the ISIT Information Theoretic Duets (along with Rajarshi Saha (currently at Amazon))
2019 – 2020 Recipient of the **B. Jayant Baliga '74 Graduate Student Fellowship Award** at RPI
2017 Recipient of the **MHRD PG Fellowship** through GATE

Invited Talks

- October 2024 **Risk-sensitive Bandits**, *IIT Bombay*, (CMinDs).
October 2024 **Risk-sensitive Bandits**, *IISc Bengaluru*, (CNI).
March 2024 **Stochastic Bandits: Complexity and Optimality**, *Imperial College London*, (Information Processing and Communications Lab).
March 2023 **Security and Safety in Multi-Armed Bandits**, *Vector Institute*, (University of Toronto).
Spring 2022 **Contaminated Best Arm Identification in Stochastic Bandits**, *Conference on Information Sciences and Systems (CISS) 2022*, (Princeton).

Technical Skills

- Languages Python, MATLAB
Libraries PyTorch, Huggingface trl

Academic Services

- 2025 Reviewer for ICML 2025
2024 Reviewer for NeurIPS 2024
2024 Reviewer for IEEE ISIT 2024
2024 Reviewer for IEEE Journal on Selected Areas in Information Theory
2023 Reviewer for AISTATS 2024
2023–2024 Reviewer for IEEE Transactions on Information Theory
2023 – 2024 Reviewer for IEEE Transactions on Mobile Computing
2023 Reviewer for IEEE ISIT 2023
2022 – 2023 Reviewer for IEEE Transactions on Communication
2022 Reviewer for AISTATS 2023
2021 Reviewer for AISTATS 2022
2021 Reviewer for AAAI 2021
2021 Reviewer for NeurIPS 2021
2019 – 2022 Reviewer for IEEE Transactions on Signal Processing

Mentoring

- Marcello Bullo**, fourth year Ph.D. student, Imperial College London.
Meltem Tatli, second year Ph.D. student, RPI.
Zirui Yan, fourth year Ph.D. student, RPI.

Referees

Prof. Ali Tajer

*Associate Professor, Department of
Electrical, Computer & Systems Engineering*
Rensselaer Polytechnic Institute
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Dr. Pin-Yu Chen

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